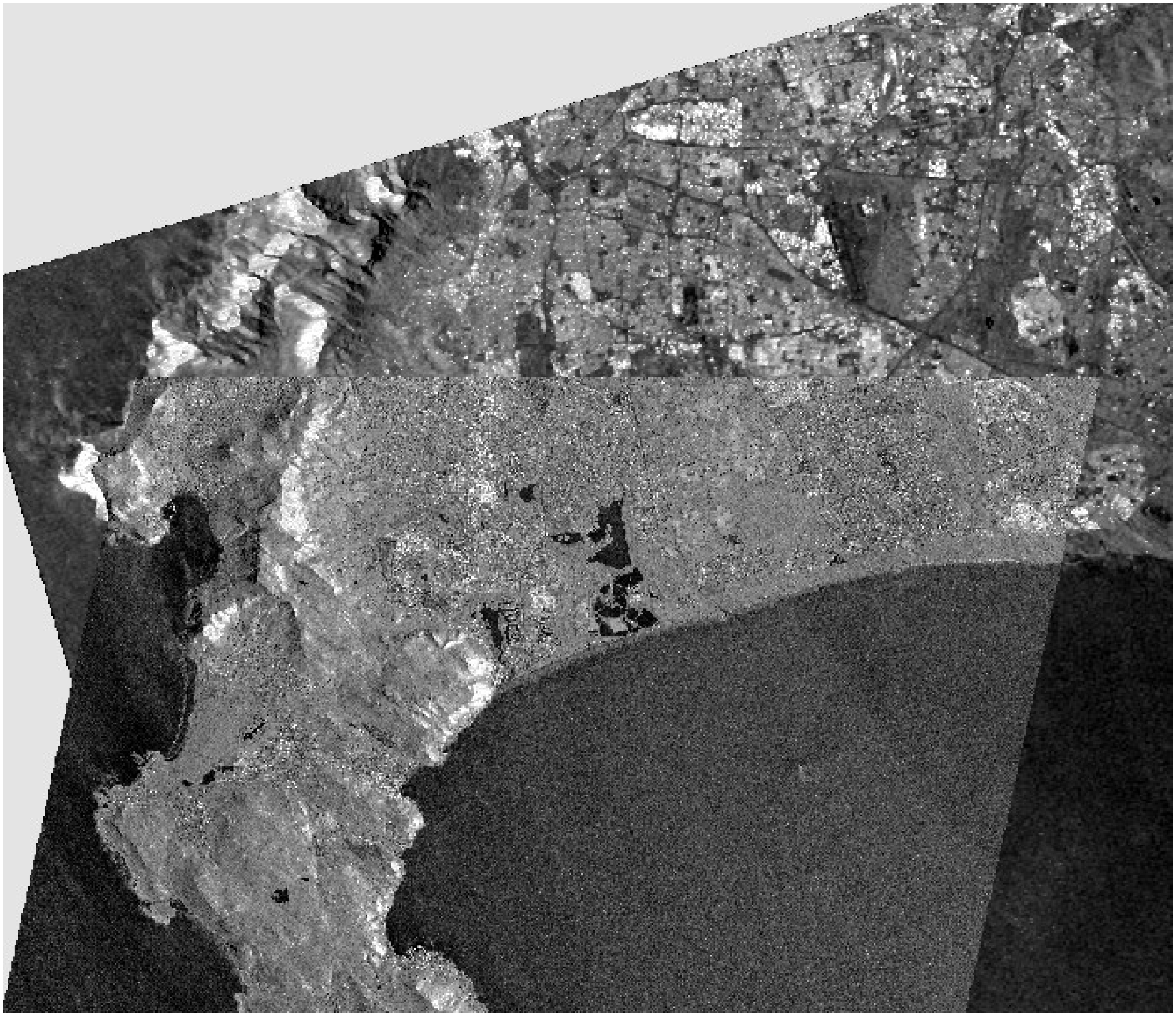

Assignment 1

A Comparison of SAR data

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Contents

Task 1

Question A

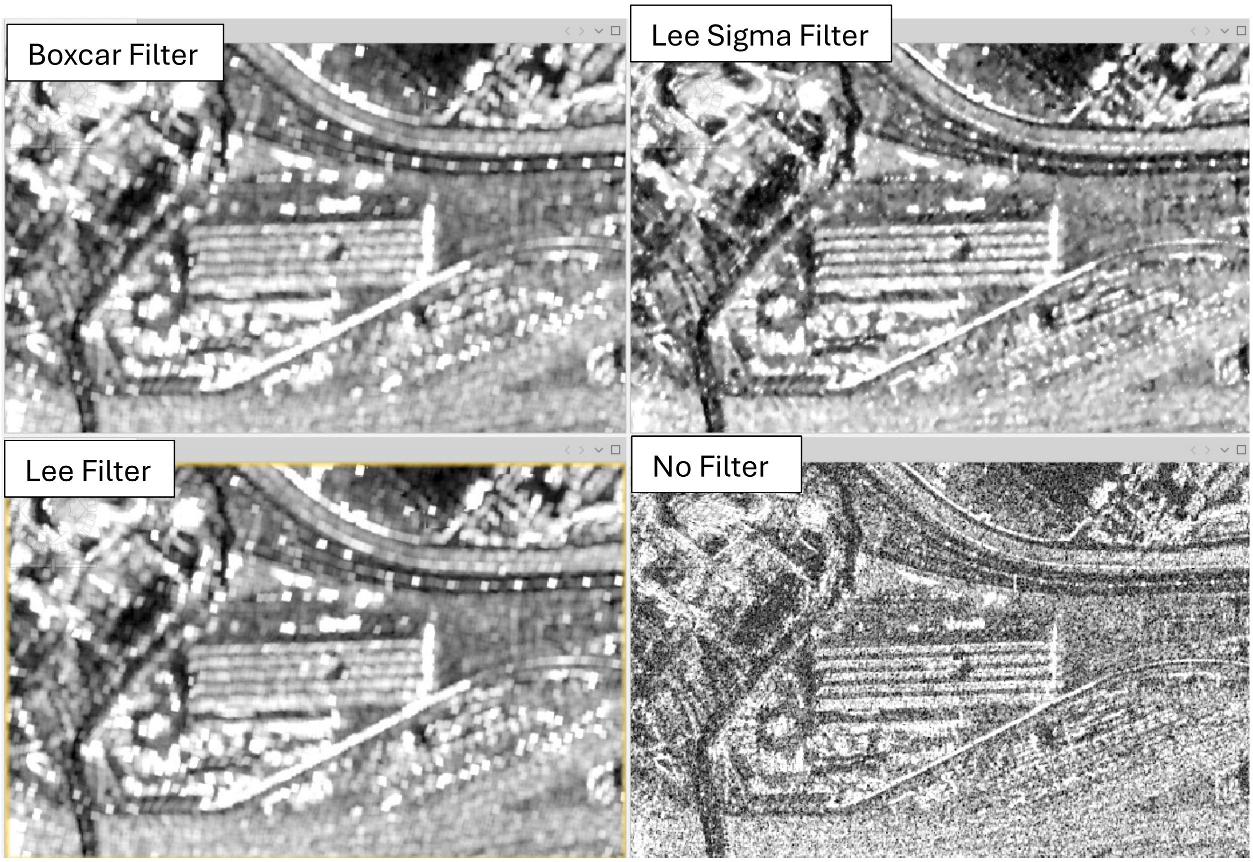

Figure 1: A comparison of different speckle filters

As seen in the figure above, when comparing the speckle filtering from the original no filter image to the filtered images it is clear that there is a significant speckle reduction in all three images. Lee Sigma still has some speckling / very bright artifacts while lee and boxcar perform the best with great speckle reduction.

Question B

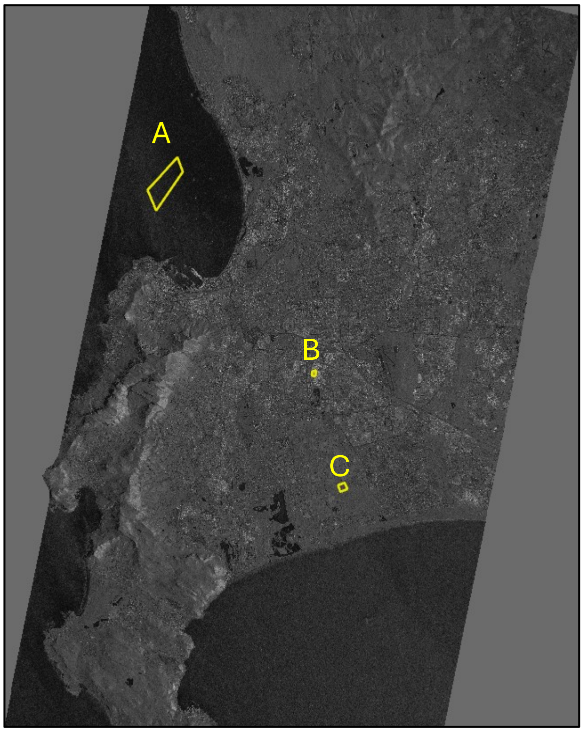

Figure 2: Study area

	Unfiltered	Boxcar	Lee Sigma	Lee
Sea	-20.7801	-17.9212	-18.3421	-17.9532
Urban	-3.60228	4.70602	1.52803	4.70602
Open land	-12.0451	-8.76734	-8.86741	-8.81951

In all three environments (sea, open land and urban), lee sigma has the smallest shift showing that it retains those original values and therefore preserves the features the best. Lee performs slightly better than Boxcar in the more homogenous environments of the sea and open land, but performs the exact same as the boxcar in highly heterogenous environment of urban.

Question C

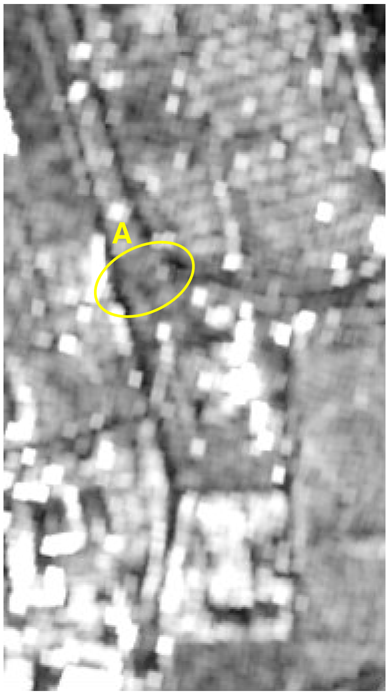

Figure 3: Boxcar line

Figure 3LEE Sigma Line

Figure 3: LEE Line

Boxcar performs the worst in the preservation of line features. As seen in figure 3, some the line features as seen in A are made indistinguishable from their surroundings.

Lee sigma performs the best in the preservation of the line features. The line features as seen in figure 3 are easily distinguishable from their surroundings.

The lee performs similarly to the boxcar as it often doesn't preserve the line features and makes it indistinguishable to its surroundings.

Question D

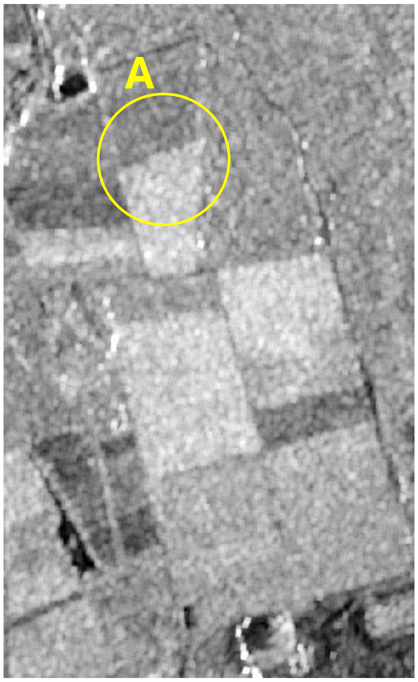

Figure 5: Boxcar Edge

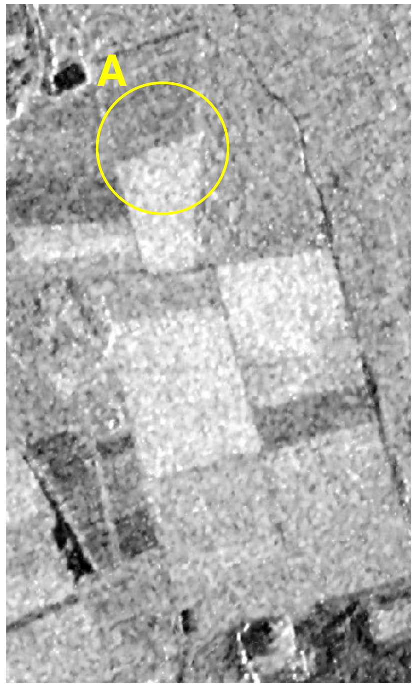

Figure 5: Lee Sigma Edge

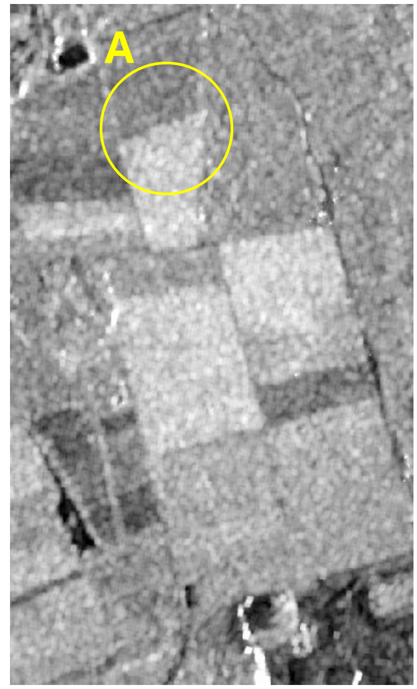

Figure 5: Lee Edge

All three filters achieve similar results. Lee sigma seems to highlight the edge differences the best followed by Lee and then Boxcar. This is due to Lee and boxcar having slightly blurry edges making the fields harder to distinguish between.

Question E

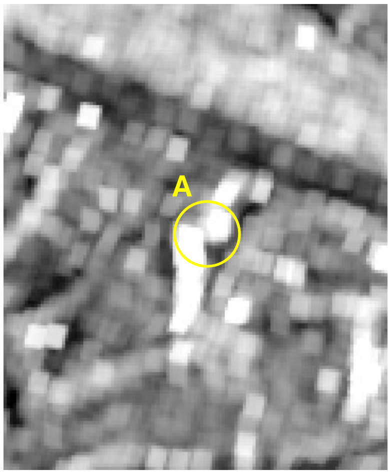

Figure 8: Boxcar point

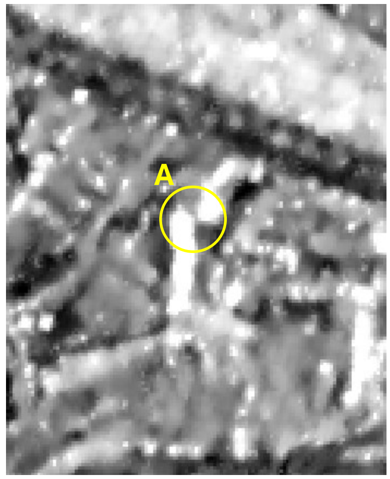

Figure 7: Lee Sigma point

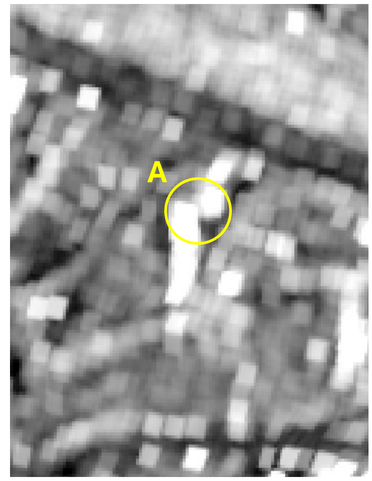

Figure 6: Lee point

The Boxcar preserves the point features the least well out of the three filters tested. It often 'smears' the point features making them less defined. This could be due to its non-adaptive mean filtering characteristics causing the bright features in A to be averaged with the surrounding low brightness features making the point features appear blurry.

The Lee Sigma clearly preserves the features the best in comparison to the other two filters. As seen in figure 7, the features A are clearly separate and therefore have been preserved well.

Lee achieves similar results to Boxcar as the two features in A in figure 8 have been merged together, but the overall brightness of the image is less than in figure 9 and therefore is slightly better than the boxcar filter.

Task 2

Speckle Reduction

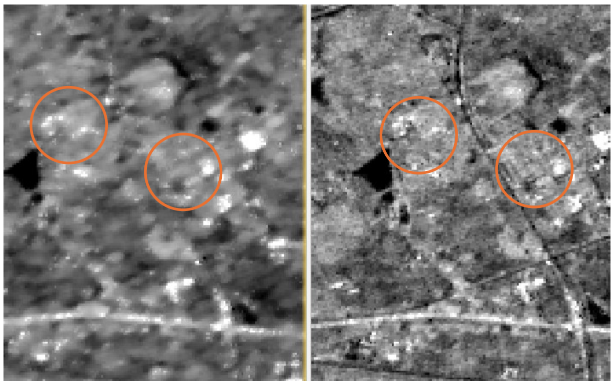

Figure 10: A Speckle content comparison of single product speckle filter (left) against Multitemporal speckle filter (right) in a rural area

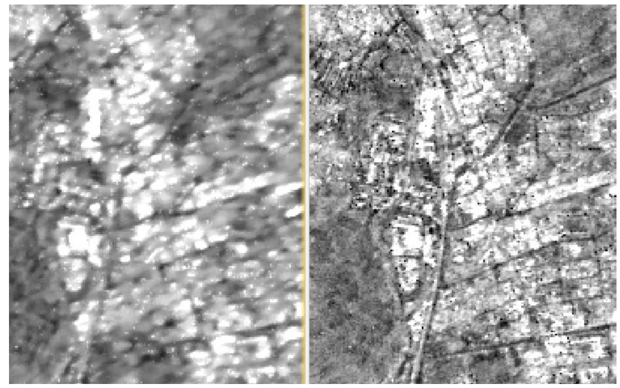

Figure 9: A Speckle Content comparison of single product speckle filter (left) against Multitemporal speckle filter (right) in an urban area

As seen above in Figure 10, there is considerably less speckling in the multitemporally filtered image than there is in the single product filtered image. This can easily be seen within the circles on Figure 10. In these circles, the circles marked by A shows a considerable amount random bright cells (aka speckling) while the circles marked B's cells are far more similar in terms of brightness.

In Figure 9, the urban areas filtered using a single product speckling filter (SPSF) shows considerable speckling than the multitemporal speckling filter (MTSF).

Mean Value Retention

	Original	Multitemporal	Single Product
Urban	-10.8065	-8.82602	-7.17887
Agriculture	-14.0968	-12.1952	-11.6361
Mountain	-4.74081	-3.43511	-1.94391

In all three land use types selected, the single product and multitemporal images are always less bright than the original. The single product filter is also always darker than the multitemporal filter image. This is due to the multitemporal filter being designed to more accurately average the brightness over time rather than the single product which only averages in spatially. The lower values will also be due to the increased speckling in the single product causing many dark cells which aren’t in the multitemporal.

Preservation of Linear Features

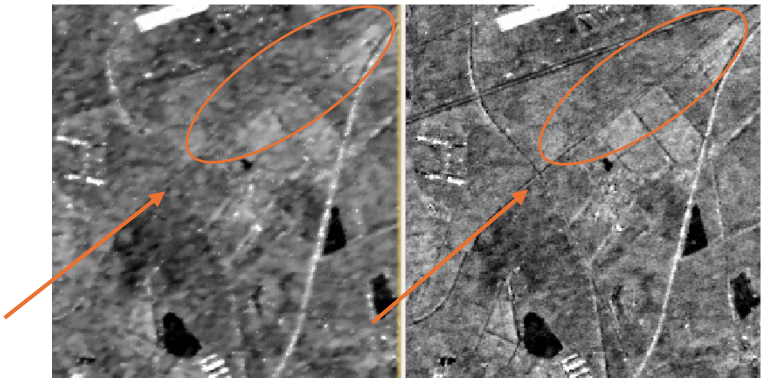
Figure 11: A comparison of single product speckle filter (left) against Multitemporal speckle filter (right) linear feature preservation quality

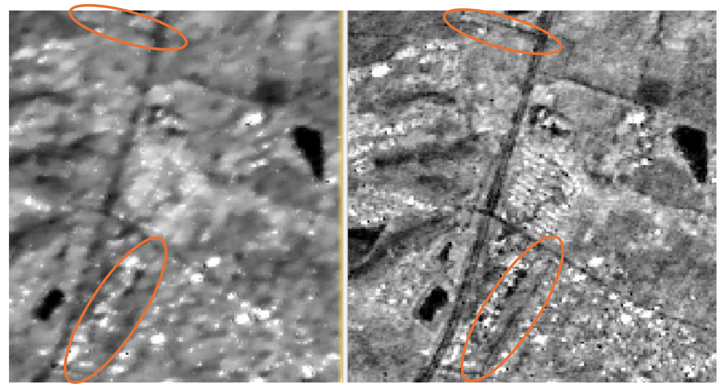
Figure 12: A comparison of single product speckle filter (left) against Multitemporal speckle filter (right) linear feature preservation quality

As seen in Figure 11, the preservation of linear features is much better in MTSF than in SPSF. It is almost impossible to view the road circled in the left image while in the image to the right it is very clear that there is a road there. The four-way stop indicated by the arrow is much harder to identify in the SPSF image than in the MTSF image. The better linear feature preservation quality of MTSF over SPSF is consistent over the whole study area as highlighted by the circles within Figure 12.

Preservation of Edge Features

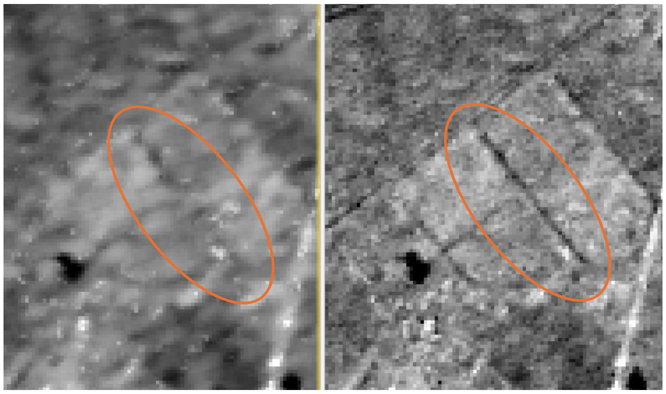

Figure 14: A comparison of single product speckle filter (left) against Multitemporal speckle filter (right) edge feature preservation quality

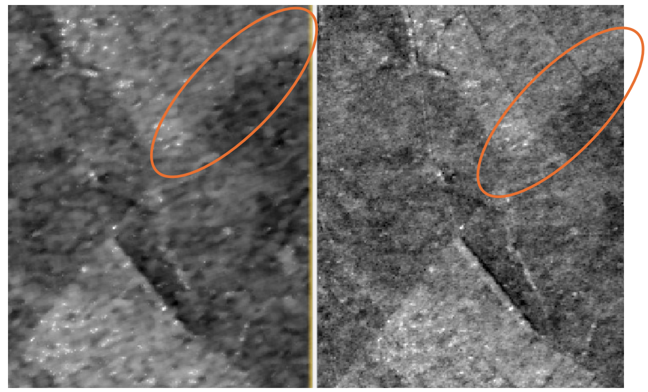

Figure 13: A comparison of single product speckle filter (left) against Multitemporal speckle filter (right) edge feature preservation quality

The MTSF clearly has better edge feature preservation quality than its SPSF counterpart. This can be seen from the features circled within Figure 13 and 14. The SPSF has a blurry effect which make it significantly more difficult to identify where one field ends and another begins. The line of trees between two fields is clearly visible in figure 6s MTSF image while in SPSF has a much less well-defined line. Between two fields of different spectral brightness, MTSF still outcompetes its SPSF counterpart. It should be noted that MTSF took significantly more data as well as processing time and power than SPSF did.

Preservation of Point Features

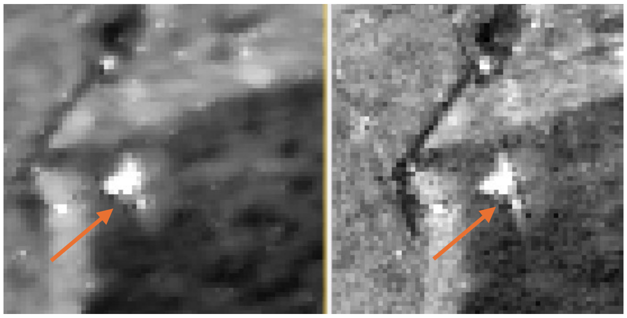

Figure 16: A comparison of single product speckle filter (left) against Multitemporal speckle filter (right) point feature preservation quality

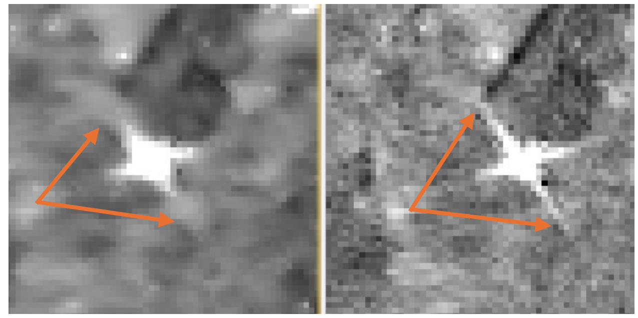

Figure 15: A comparison of single product speckle filter (left) against Multitemporal speckle filter (right) point feature preservation quality (Corner Reflector)

As seen in Figure 16, the point features are of fairly similar quality between the multitemporal and single product filter. The MTSF is slightly better than the SPSF as the point feature in the SPSF has slight blurring on its left-hand side indicated by the orange arrow.

The SPSF is significantly better than the MTSF at minimizing the effects of corner reflectors. This can be clearly seen in Figure 15 where the ‘tails’ of the corner reflector are significantly longer in the MTSF than in the SPSF.

Task 3

Question 1:

Water

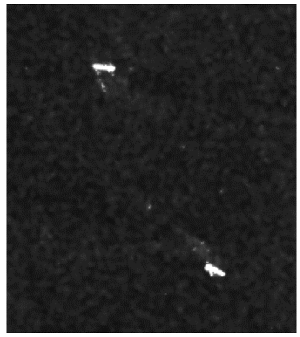

Figure 17: Double-bounce off ships in the water VV

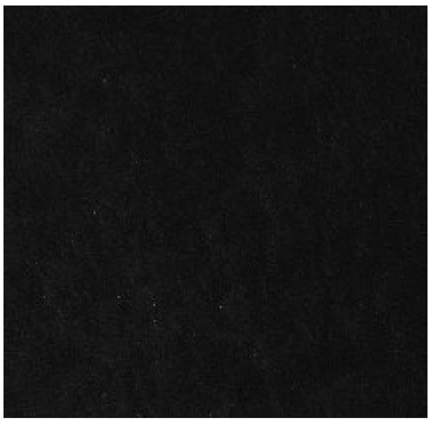

Figure 19: The ocean after processing VV

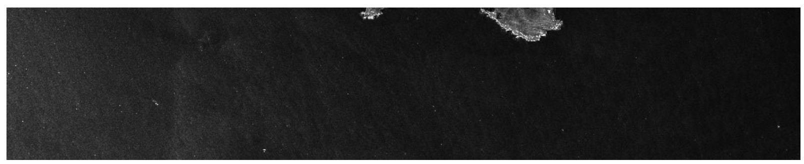

Figure 18: Image showing the grey to black gradient of the water body VV

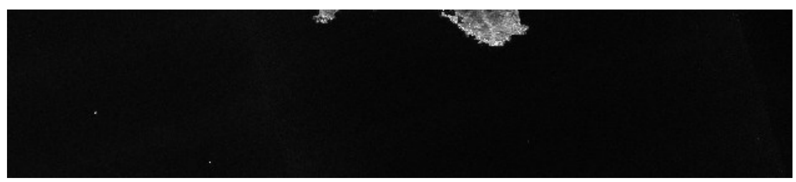

Figure 20: VH polarization colour gradient over a large body of water

As seen Figure 19, the water appears very dark. This is due to the specular reflection associated with smooth surfaces such as water. This type of reflections causes there to be almost no backscatter. There areas especially in the south western areas (shown in Figure 18) where the water appears a slightly grey colour most likely due to the Bragg scattering caused by waves / rough water. The grey appearance of the water could also be due to the steeper incidence to the sensor as the image was take from the west. This is expected as VV polarized imagery is highly sensitive to surface scattering on smooth surfaces. Figure 17 shows an example of corner reflectors in the water which create bright spots due to double-bounce, these are most likely ships. This double bounce causes lots of backscatter back to the sensor and therefore appears very bright.

VH polarisation acts similarly in many cases as there is also minimal backscatter from the water, but there are some key differences. Firstly, VH polarisation is not as sensitive to rough surfaces as VV is. This can clearly be seen when comparing figures 2 and 4, where there is no gradient in 4 but there is in figure 2.

Urban Areas

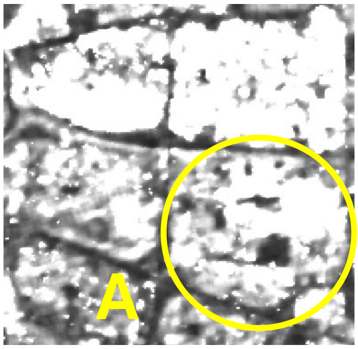

Figure 22: Backscatter of Urban Areas from VV polarisation

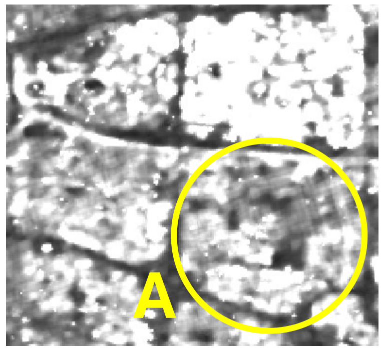

Figure 21: Backscatter of Urban Areas from VH polarisation

The urban areas are significantly brighter than other parts of the image visible in Figure 22. This is due to the backscattering and corner reflection associated with urban areas creating high backscatter levels. These are expounded due to the nature of VV polarization on vertical features making these areas brighter than what you would expect on an HH or VH polarized image. The increased backscatter due to the VV polarisation becomes apparent when comparing it to the VH polarisation. As seen in Figure 21and Figure 22 in circle A, the backscatter from the VV polarisation is dramatically more than the polarization in the VH polarised image.

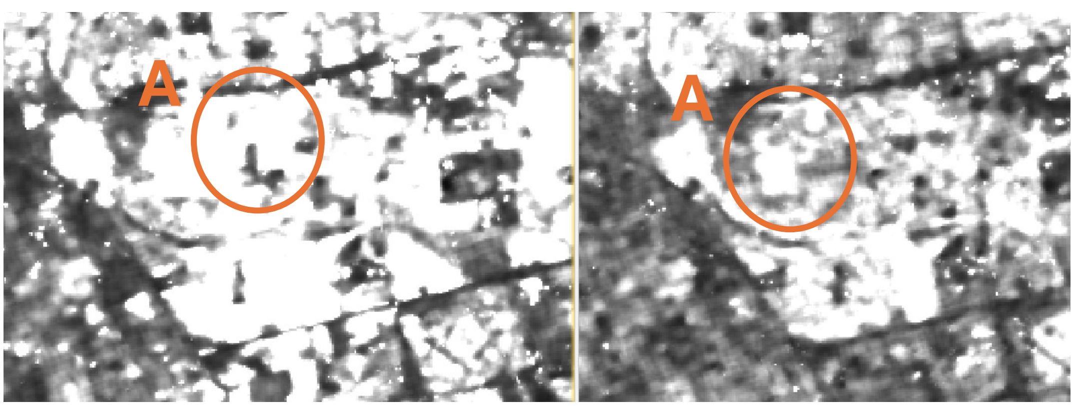

Figure 23: VV (right) and VH (Left) polarisation corner reflectors

As seen in Figure 23, VV polarisation receives direct strong double bounce backscattering. This causes a blooming or blurring effect as well as generates many corner reflectors. These effects combine causes point features to become indistinguishable from each other.

VH on the other hand receives fairly depolarised signals which are therefore much weaker and less saturated than those received in VV polarised imagery. This causes less blurring and less corner reflectors created. This means that the VH as seen in Figure 23, has more defined point features within the busy urban environment than the VV does.

Agriculture

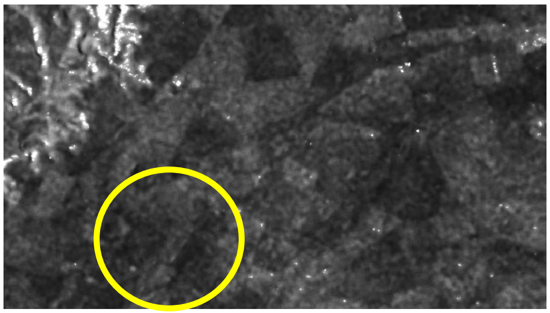

Figure 25: VV polarized image of an agricultural field

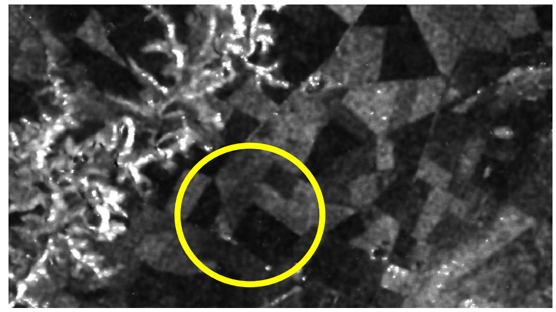

Figure 24: VH polarized image of an agricultural field

The microwaves scatter in multiple directions when it comes into contact with vegetation or live crops. This depolarizes the energy, allowing for easy detection from the VH band. These live vegetation or crops will appear significantly brighter than the bare soil or ploughed fields which exhibit clear specular reflection due to its smooth characteristic.

In contrast to the VH band, the VV band struggles to detect depolarised microwaves. This means that the specular reflection off the bare/ploughed fields looks very similar to the depolarised waves from the live crops or vegetation. This makes distinguishing bare and live fields very difficult, while the VH band shows significant contrast due to its ability to detect depolarised waves. This can be clearly seen when comparing A in Figure 24 and Figure 25. This is why VH bands are often used to detect fields and live vegetation.

Steep Terrain

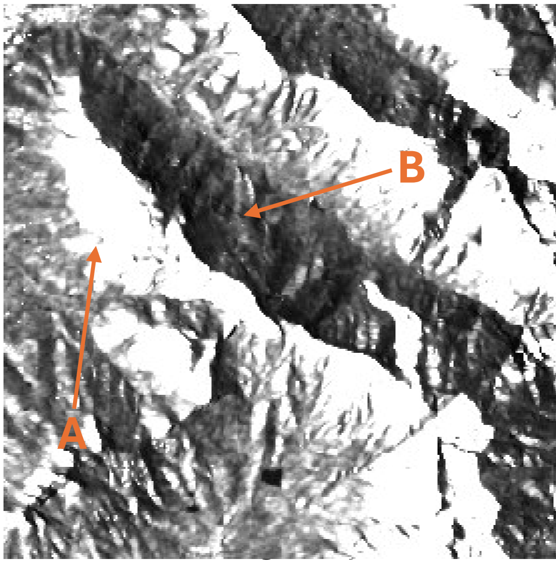

Figure 26: VV polarisation image of steep terrain

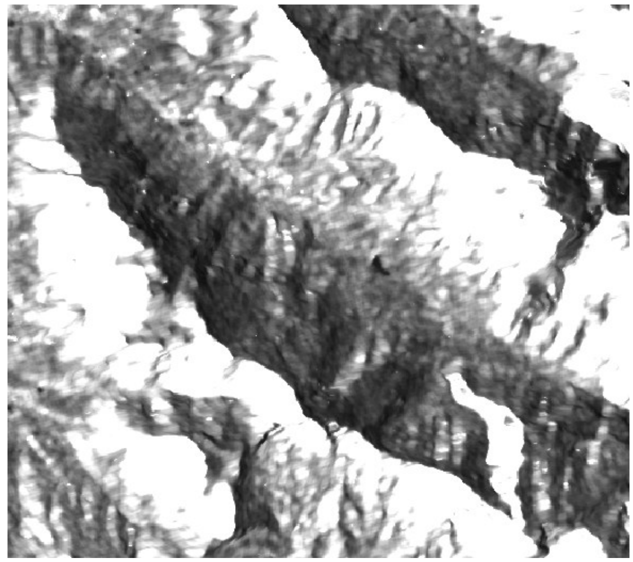

Figure 27: VH polarisation image of steep terrain

When comparing A and B on Figure 26, to A and B in Figure 27 we can see that they are very similar. This is due to the extreme terrain which reflects both a polarised and depolarised wave. Although it should be noted that the VHs shadows are not as dark as those in VV.

Roads and Railways

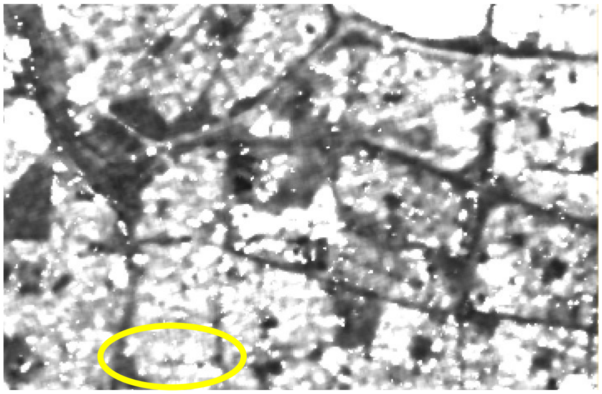

Figure 29: VV polarized image showing roads within an urban area

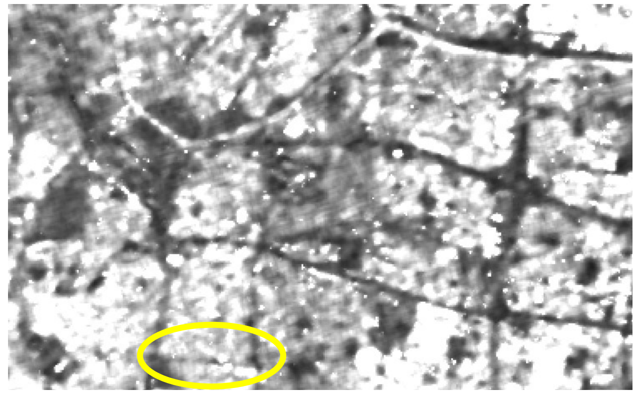

Figure 29: VH polarized image showing roads within an urban area

The large roads and railways are smooth surfaces so therefore the VV and VH represent these features fairly similarly. Although due to the urban environment being better represented in VH, smaller roads are more visible in this polarisation than in VV polarization. This is due to the corner reflectors and strong double bounce backscatter making the buildings appear larger and therefore hide the smaller roads. This difference between the two polarisation bands is clearly visible in the circles in Figure 29 and Figure 29.

Unexplainable Phenomenon

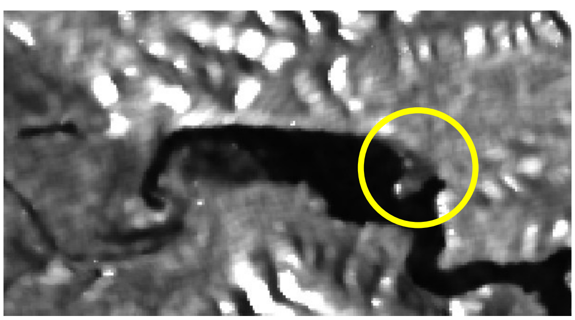

Figure 31: VV polarized image of a river

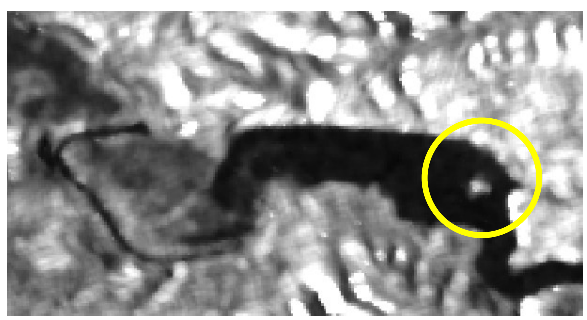

Figure 30: VH polarized image of a river

As seen circled in Figure 31 there appears to be a spit, but in Figure 30 this spit appears to be an island as it is no longer connected to the land. This is very strange and should be

investigated.

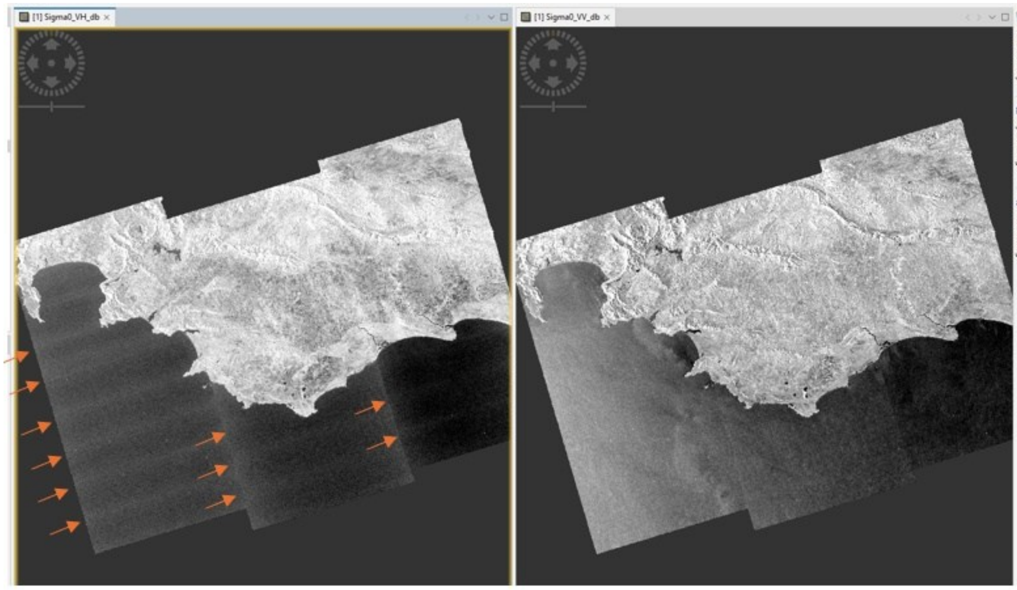

Figure 32: Investigation into strange lines on the VH image

In the VH image on the left, clear lines can be seen traveling from left to right indicated by the arrow and circle. This is interesting as only north to south lines should be visible from merging images together. Otherwise if these horizontal lines are from scanning back and forth why are they only visible in the VH band?

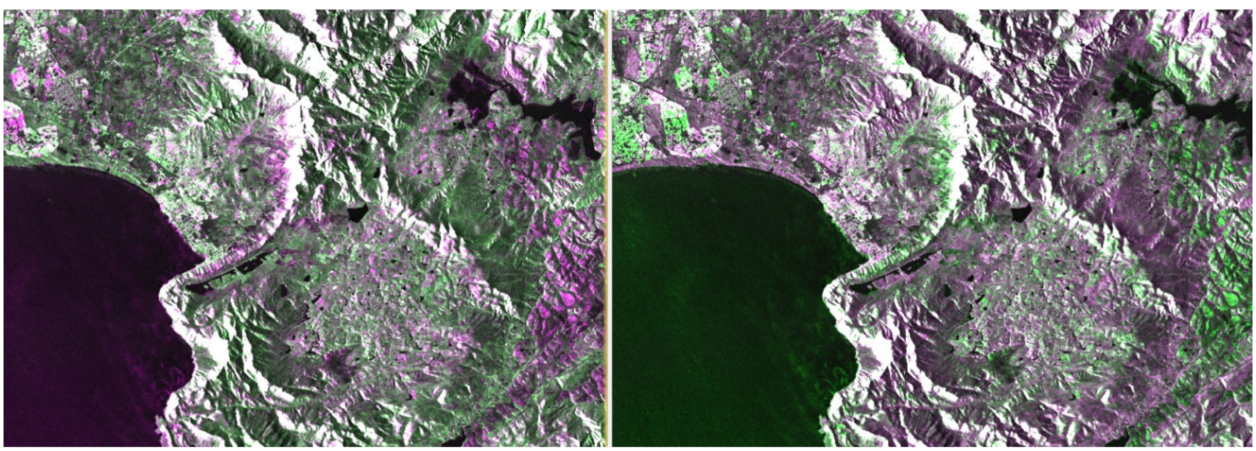

Figure 33: False coloured SAR image

Experimentation was done false colouring various bands. The most useful was the combination above. This was achieved by assigning blue and red to VV bands and green to VH band. This causes vegetative areas to appear green, while rougher areas such as bare soil and recently ploughed agricultural fields to appear purple. The sea on the west side of the image also has a purple hue probably due to the sea being rougher in this area.

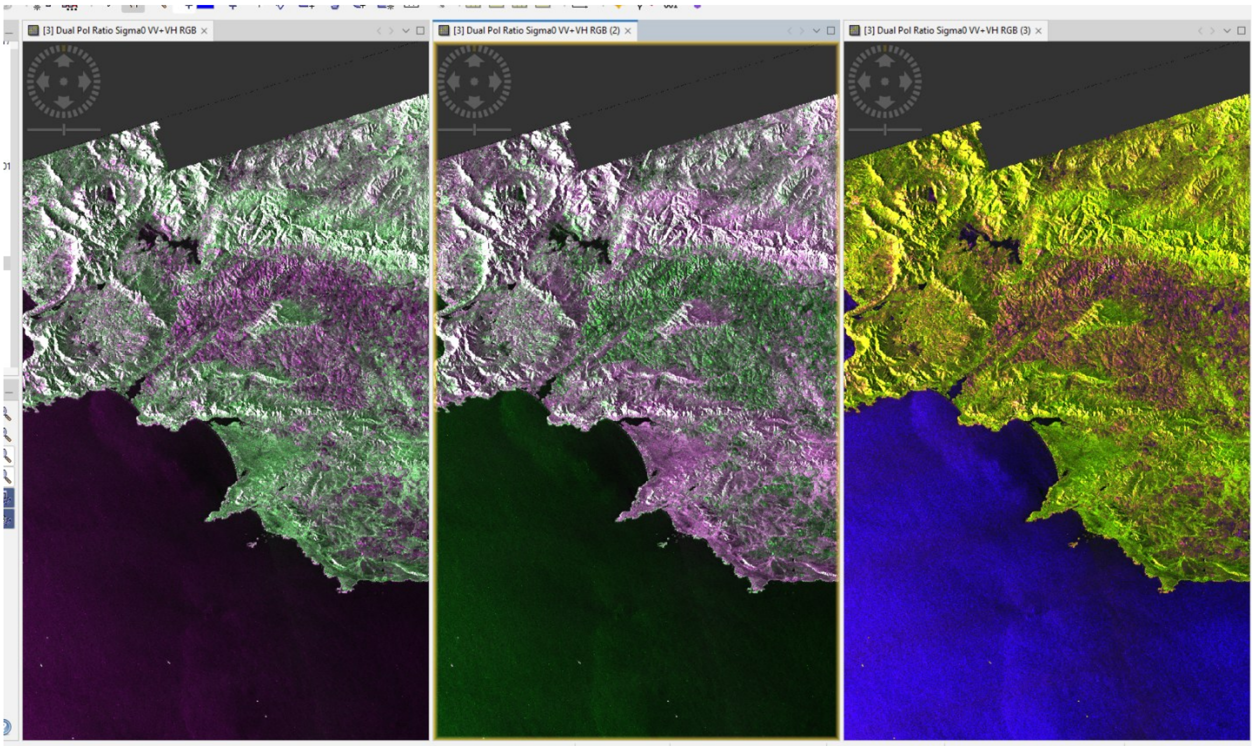

Figure 34: Another false colour map

All these maps are false coloured sentinel-1 images, showing the same landcovers as Figure 34, but in different colours. The blue and yellow colours of the third map may have better potential to show these landcovers to red-green colourblind people.

Task 4

Comparative Validity

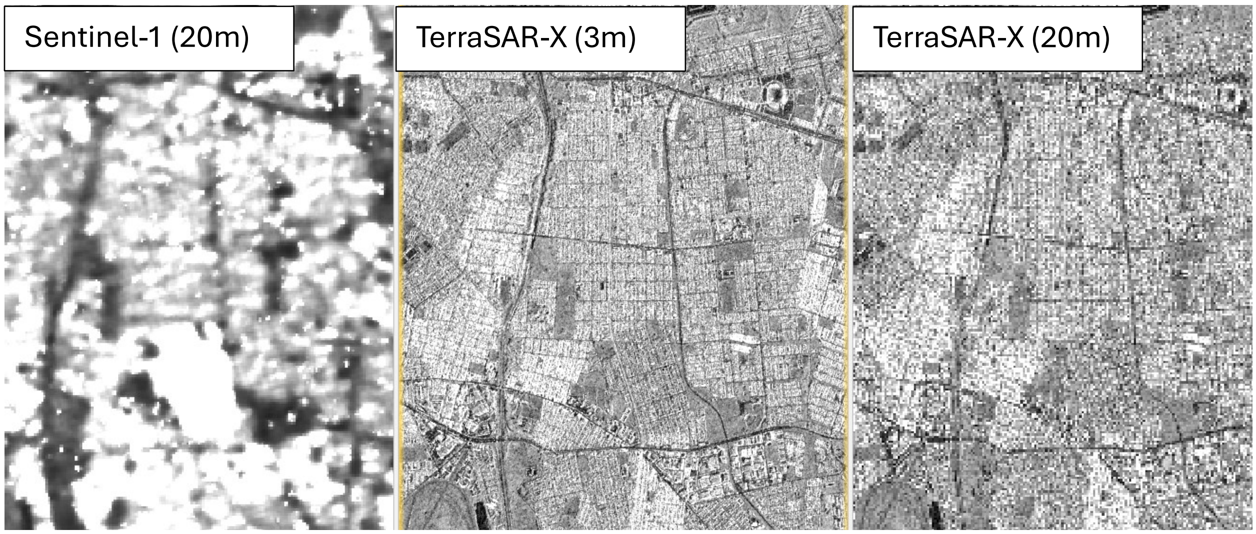

Figure 35: A comparison of the different satellites at different resolutions

For this task, A Sentinel-1 and TerraSAR-X SAR images are being used to compare the backscatter patterns of C and X band images respectively. In order to compare these two images, it must be ensured that they have the same processing properties and have been manipulated in the same way so that the differences and similarities identified can be attributed to the band difference and not other factors.

The spatial resolution is a key issue as smaller resolutions greatly amplify the size of corner reflectors and the double bounce backscatter. Both images were terrain corrected to a pixel spacing or resolution of 20m.

The stretch type and the range were also kept consistent. Both were restretched using a maximum-minimum stretching type. The range was determined by taking the lower and higher average of combination of the images; therefore, the range was changed to 34 to -44.5.

Polarization

The different polarization types between these images should also be considered as this heavily affects the backscatter. The TerraSAR-X image is HH polarized and will therefore be highly sensitive to rough surfaces so any amount of ‘roughness’ will be visible on this image. Vertical objects on the surface such as buildings in urban areas or very steep terrain will cause very strong polarized signals.

In contrast the Sentinel-1 has VV and VH polarized bands. The VV polarized band will be highly sensitive to smooth surfaces due to its sensitivity to bragg scattering. It also picks up strong signals from vertical structures, but unlike HH, it lacks a lot of double bounce backscatter. This means that it should be expected that moisture variations and water roughness variations will be easily visible within this band as seen below Figure 36.

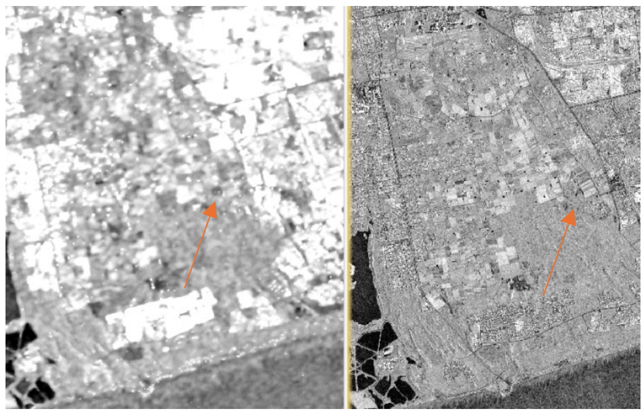

Figure 36: A comparison of agricultural fields between VV polarized (left) and HH polarized (right)

The VH band is sensitive to volume scattering. This is due to the signal emitted from the sensor reflecting multiple objects before returning to the sensor causing depolarized returns. These cross polarized returns are often due to thick foliage/vegetation. Therefore, the VH band will highlight vegetation and live crops more than the same polarized VV or HH images. It should be noted that the attenuation of the polarization is dependent on the vegetation biomass as well as the wavelength and incidence angle of the emitted signal.

Wavelength

Radar waves always interact the most with objects /features of a similar size to their wavelength. This means that waves with smaller wavelengths will interact with smaller features such as leaves and twigs on trees while waves with larger wavelengths will only interact with larger objects such as the trunk or ground below the leaves/vegetation as seen in Figure 37.

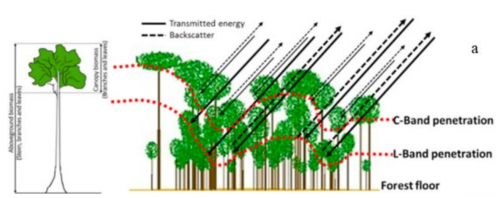

Figure 37: Penetration of different wavelengths

The Sentinel-1 sensor captures SAR images in C-band while TerraSAR-X captures images in X-band. This means that TerraSAR-X will interact with more features and be more sensitive to rough surfaces than the sentinel-1 will be. As seen below in Figure 38, This means that even though they are both in the same scale, the TerraSAR-X will show more detail than the Sentinel-1 images. Therefore, the TSX shows more backscatter within the natural vegetation and live agriculture, as well as small roughness on water surfaces such as the ocean than the Sentinel-1. The sentinel-1 will penetrate through the vegetation causing slightly less reflection, although due to the cross polarization that will occur, the VH polarized sentinel-1 shows similar backscatter intensity to the TSX.

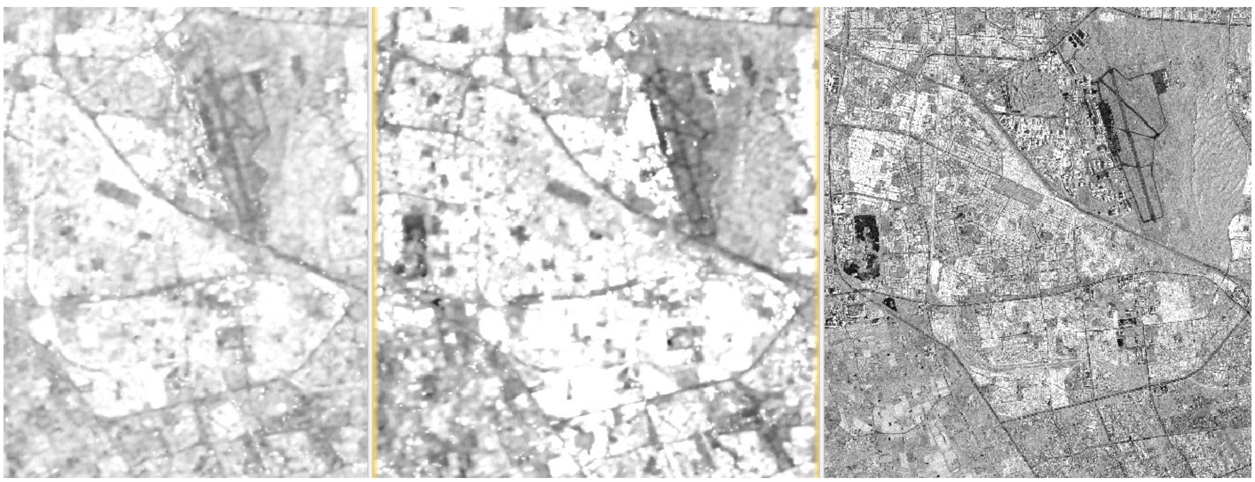

Figure 38: A comparison of the level of detail between the sentinel and TerraSAR-X satellites

Time of day

The TSX1 and Sentinel1 images were both taken on the same day, but taken at different times of the day. The TSX1 was taken in the morning at approx. am while the sentinel1 image was taken in the evening at approx. 5:30 pm. The time of day will affect the moisture of the soil and vegetation. Due to TSX1s small X band, this increased soil moisture will make the double bounce affect stronger as there will be much less soil penetration. The vegetation will also have more water on it as well as in its structure and therefore there will be more backscatter. In contrast the sentinel-1 image was taken in the afternoon, this means that the plants and soil are drier and therefore the same field in TSX1 will appear brighter than that in the Sentinel as well as there will be less volume scattering from the vegetation making the TSX and sentinel images more similar in respect to the reflection. This higher backscatter phenomenon when soil or vegetation is wetter is due to the waters strong dielectric properties.

Look Direction

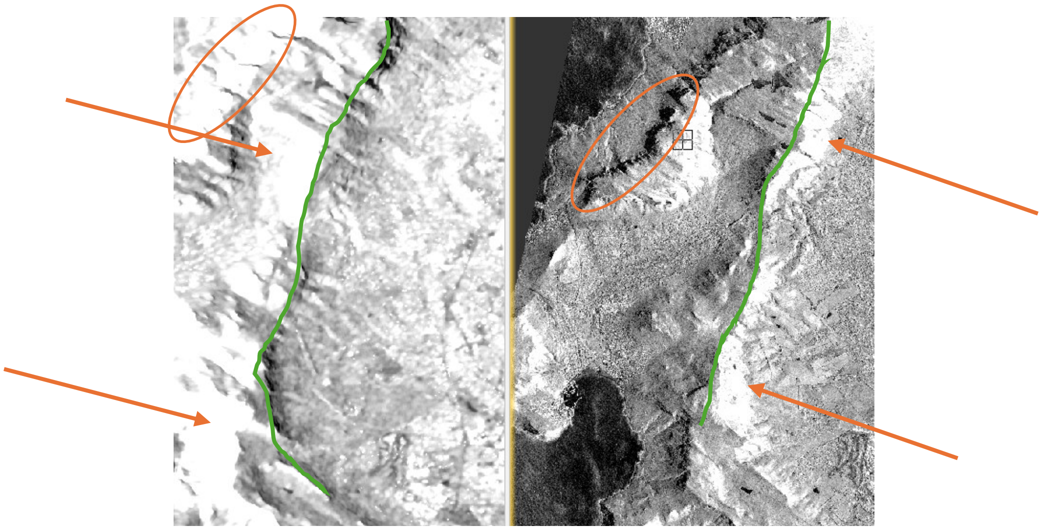

Figure 39: A comparison between the look direction of the Sentinel-1 and TerraSAR-X satellites.

The look direction of the satellite determines the direction of the three side-looking geometric distortions of SAR satellite data, namely, radar foreshortening, layover and shadow. The green line on Figure 39 shows the ridge of the mountain, with the arrows indicating the look direction causing a bright reflection on the mountain, the circles highlight the difference in shadow created on the mountain. The TSX1 is on a descending pass while the Sentinel-1 is on an ascending pass. This means that the look directions of these satellites will be the opposite, with the TSX1 looking west and Sentinel-1 looking east. This means that the shadows which always cast away from the sensor will be facing opposite directions. The foreshortening and layover will also be in different directions so this must be taken into account when analyzing the image.

The backscatter will also appear on opposite sides of vertical/high objects and features such as mountains and buildings. This means that different aspects of the same mountain can be analyzed. For example, the east side of the mountain is in shadow in the TSX and therefore can’t be false colored, etc. but the west side can be classified and analyzed and vice versa for the sentinel-1.

Incidence Angle

The incidence angle is very important as it greatly affects the geometric distortions as well as the backscatter patterns in an image. Steeper incidence angles create greater degrees of geometric distortion in combination with steep, high features such as mountains.

The backscatter is higher as the angle of the incidence angle increases, meaning that the closer to the antenna any of point the image is, the higher the backscatter will be in comparison to a cell of a similar environment.

References

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